Cybersecurity - Graduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is an authentication method?
   (a) Secret question
   (b) Biometric
   (c) SMS code
   (d) All of the above

2. A hash function guarantees the integrity of a message. It guarantees that the message has not be
   (a) Replaced
   (b) Overview
   (c) Changed
   (d) Violated

3. The AH Protocol provides source authentication and data integrity, but not
   (a) Integrity
   (b) Privacy
   (c) Nonrepudiation
   (d) Both A & C

4. Let I = (S,V)I=(S,V) be a MAC. Suppose S(k,m)S(k,m) is always 5 bits long. Can this MAC be secure?
   (a) No, an attacker can simply guess the tag for messages
   (b) It depends on the details of the MAC
   (c) Yes, the attacker cannot generate a valid tag for any message
   (d) Yes, the PRG is pseudorandom

5. ______________________ is the central node of 802.11 wireless operations.
   (a) WPA
   (b) Access Point
   (c) WAP
   (d) Access Port

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key can be reused multiple times for different sessions between the same parties.
7. True or False: A buffer overflow occurs when a pointer is used to access memory not allocated to it, leading to potential security vulnerabilities.
8. True or False: Weak or non-existent authentication mechanisms are not issues associated with the presentation layer in cybersecurity.
9. True or False: The message must be encrypted and decrypted at the sender site to maintain confidentiality during transmission.
10. True or False: A proxy firewall operates primarily at the application layer, filtering traffic based on application-specific protocols.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of message integrity in cybersecurity, analyzing its significance in ensuring that data arrives at the receiver exactly as it was sent.
12. Discuss the role of the session layer in the OSI model and analyze how failed session attempts can facilitate brute-force attacks on access credentials in cybersecurity.

End of Paper